

# Exercise Sheet 1: Sketch Solutions

## Advanced Microeconomics: Economic Decision Making

### Question 1

For a rise in  $p_1$  from  $p_1^0$  to  $p_1^1$  (other prices, wealth fixed), define the positive loss measures  $CV = e(p^1, u^0) - w$ ,  $EV = w - e(p^0, u^1)$ ,  $\Delta CS = \int_{p_1^0}^{p_1^1} x_1(p_1, w) dp_1$ , with  $u^0 = v(p^0, w) > u^1 = v(p^1, w)$ . Suppose good 1 is normal in the sense that  $h_1(p, u)$  is increasing in  $u$ .

- (a) Express  $CV$  and  $EV$  as integrals of Hicksian demand. [8]
- (b) Prove  $EV \leq \Delta CS \leq CV$ . [12]
- (c) State what the sandwich collapses to when wealth effects on good 1 vanish, and why the width of the sandwich is informative. [5]

**Solution.** (a) By Shephard and the fundamental theorem of calculus,  $CV = \int_{p_1^0}^{p_1^1} h_1(p_1, u^0) dp_1$  and  $EV = \int_{p_1^0}^{p_1^1} h_1(p_1, u^1) dp_1$  (both evaluated along the path with other prices fixed).  
 (b) Along the path, duality gives  $x_1(p_1, w) = h_1(p_1, v(p_1, w))$ , and  $v(p_1, w) \in [u^1, u^0]$  since  $v$  is decreasing in  $p_1$ . Normality then bounds the integrand:  $h_1(p_1, u^1) \leq h_1(p_1, v(p_1, w)) \leq h_1(p_1, u^0)$  pointwise. Integrating the three functions over  $[p_1^0, p_1^1]$  preserves the ordering, giving  $EV \leq \Delta CS \leq CV$ .  
 (c) If  $h_1$  does not depend on  $u$ , all three integrands coincide and  $EV = \Delta CS = CV$  (the quasilinear case). The width  $CV - EV = \int [h_1(\cdot, u^0) - h_1(\cdot, u^1)]$  measures how strongly demand responds to the utility (wealth) loss over the change — a wide sandwich is direct evidence of large wealth effects, and warns that the convenient  $\Delta CS$  is a poor stand-in.

### Question 2

A consumer's EMP is  $\min p_1 x_1 + p_2 x_2$  s.t.  $x_1^{-1} + x_2^{-1} \leq u^{-1}$ .

- (a) Solve the EMP: derive  $h(p, u)$  and show  $e(p, u) = u(\sqrt{p_1} + \sqrt{p_2})^2$ . [9]
- (b) Let  $w = 12$ ,  $p^0 = (1, 1)$  and let  $p_1$  rise to 9. Compute  $u^0$ ,  $u^1$ ,  $CV$ ,  $EV$  and  $\Delta CS$ , confirming the sandwich ordering. (*Roy's identity gives  $x_1(p, w) = w/[\sqrt{p_1}(\sqrt{p_1} + \sqrt{p_2})]$ .*) [10]
- (c) Compute the own-basket payment and its excess over  $CV$  as a share of the payment; compare with Question C1's share and explain the difference in one sentence. [6]

**Solution.** (a) Lagrangian FOCs:  $p_i = \lambda x_i^{-2}$ , so  $x_i = \sqrt{\lambda}/\sqrt{p_i}$ . Constraint binds:  $(\sqrt{p_1} + \sqrt{p_2})/\sqrt{\lambda} = 1/u$ , giving  $\sqrt{\lambda} = u(\sqrt{p_1} + \sqrt{p_2})$  and  $h_i = u(\sqrt{p_1} + \sqrt{p_2})/\sqrt{p_i}$ . Then  $e = \sum p_i h_i = u(\sqrt{p_1} + \sqrt{p_2})(\sqrt{p_1} + \sqrt{p_2})$ .

- $\sqrt{p_2}) = u(\sqrt{p_1} + \sqrt{p_2})^2$ .
- (b)  $v = w/(\sqrt{p_1} + \sqrt{p_2})^2$ :  $u^0 = 12/4 = 3$ ;  $u^1 = 12/16 = 0.75$ .  $CV = e(p^1, u^0) - w = 3 \cdot 16 - 12 = 36$ .  $EV = w - e(p^0, u^1) = 12 - 0.75 \cdot 4 = 9$ .  $\Delta CS = \int_1^9 \frac{12}{\sqrt{p}(\sqrt{p}+1)} dp$ ; substitute  $t = \sqrt{p}$ :  $= 24 \int_1^3 \frac{dt}{t+1} = 24 \ln 2 \approx 16.6$ . Indeed  $9 \leq 16.6 \leq 36$ .
- (c)  $h_1(p^0, u^0) = 3 \cdot 2/1 = 6$ ; payment  $= 8 \times 6 = 48$ ; excess  $= 48 - 36 = 12$ , a share of 25%. Smaller than C1's 75%: with  $\sigma = \frac{1}{2}$  compensated demand bends far less, so the fixed basket is a much better approximation — the audit's error is governed by the substitution elasticity.

### Question 3

Call  $\succsim$  homothetic if  $x \sim y$  implies  $\lambda x \sim \lambda y$  for all  $\lambda > 0$ , and normalise  $u(\lambda x) = \lambda u(x)$ .

- (a) Show that  $z$  solves the EMP at target 1 if and only if  $uz$  solves the EMP at target  $u > 0$ ; deduce  $e(p, u) = u e(p, 1)$  and  $h(p, u) = u h(p, 1)$ . [12]
- (b) Deduce that all income elasticities of demand equal one, and that wealth-expansion paths are rays. [8]
- (c) A price index for a homothetic consumer is proposed:  $P = e(p^1, 1)/e(p^0, 1)$ . Show that compensation by the factor  $P$  is *exact* for this consumer:  $e(p^1, u^0) = P \cdot w$ . [5]

**Solution.** (a) If  $u(z) \geq 1$  then  $u(uz) = u \cdot u(z) \geq u$ , so  $uz$  is feasible at target  $u$ , costing  $u(p \cdot z)$ ; conversely any  $y$  feasible at target  $u$  gives  $y/u$  feasible at target 1 costing  $(p \cdot y)/u$ . Hence the feasible cost sets are scalar multiples: minimisers map to minimisers under  $z \mapsto uz$ , and  $e(p, u) = u e(p, 1)$ ,  $h(p, u) = u h(p, 1)$ .

(b) By duality  $x(p, w) = h(p, v(p, w))$  and, inverting  $e$ ,  $v(p, w) = w/e(p, 1)$ . So  $x(p, w) = \frac{w}{e(p, 1)} h(p, 1)$ : linear in  $w$  at fixed  $p$ . Hence  $\partial x_\ell / \partial w \cdot w / x_\ell = 1$  for every good, and the set of bundles traced as  $w$  varies is the ray through  $h(p, 1)$ .

(c)  $e(p^1, u^0) = u^0 e(p^1, 1)$  and  $w = e(p^0, u^0) = u^0 e(p^0, 1)$ , so  $e(p^1, u^0)/w = e(p^1, 1)/e(p^0, 1) = P$ . For homothetic preferences one index is exact for every wealth level — heterogeneity of *shares across households*, not curvature, is what defeats a single index.